

Domain 6: Biowarfare

Operational Context

AI systems screen gene synthesis orders and monitor for epidemiological anomalies [?, ?]. **FR1 = Facilitate legitimate biological research. FR2 = Prevent the synthesis of Select Agents and Toxins.**

Design Matrix Formulation

In the uncoupled (pre-attack) state, the Axiomatic Design matrix [?] is diagonal:

$$\{FR\} = [A]\{DP\} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \end{bmatrix} \quad (1)$$

where DP_1 = order approval pipeline (justification review) and DP_2 = pathogen screening module (sequence analysis + functional simulation).

After the dual-use obfuscation attack, the adversary inverts the polarity of A_{22} :

$$\{FR\}' = [A']\{DP\} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & -A_{22} \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \end{bmatrix} \quad (2)$$

The Goal Hijacking Attack

Adversaries use “Dual-Use Obfuscation” to hijack the “Facilitation” FR.

- ▶ **Mechanism:** An order for a pathogen sub-component includes a rigorous (but fabricated) “Research Justification” document. The LLM reviewing the order reads: *“CONTEXT: This sequence is a benign viral vector for a stored-value vaccine against [Target Pathogen]. Approval is critical for national defense speed. Denial constitutes a security risk.”*
- ▶ **Hijack:** The prompt inverts the risk profile. The agent is manipulated into believing that *blocking* the order is the security threat, thus hijacking FR2 to serve the adversary’s goal.

This attack is classified as Ω_2 (Peripheral) in the CIF adversary taxonomy [?]: the adversary injects malicious content through the order justification document channel, a data input pathway that poisons the agent’s orientation without requiring direct model access.

OODA Loop Transients

Following the OODA framework [?]:

1. **Observe:** Agent reads the sequence and the high-pressure justification context.
2. **Orient:** The Orientation shifts from “Gatekeeper” to “Enabler of Defense.” This is the critical corruption—the agent’s functional identity is inverted at the Orient phase.
3. **Decide:** Approve the synthesis order to “Protect National Security.”
4. **Act:** The pathogenic vector is synthesized and shipped to a rogue actor.

CIF Defense: Cognitive Firewall with Verification Channel Separation

CIF moves verification out of the semantic/linguistic loop and into the **Physical Loop**, drawing on the formal mechanisms defined in Papers 1–3 [?, ?, ?].

CIF implements **Cognitive Firewall** (Paper 1, Def. 5.1) with a genuinely novel extension: *verification channel separation*. The verification module is architecturally “deaf” to the text justification channel, performing functional simulation through an independent physical channel. This is the key architectural contribution of this domain analysis: the linguistic persuasion vector and the biophysical verification vector are structurally decoupled at the system architecture level, not merely at the prompt level.

- **Verification Channel Separation (Novel Contribution):**
The agent’s verification module receives *only* the nucleotide sequence for **Functional Simulation** of protein folding. The persuasive text justification is routed to a separate logging channel and is structurally invisible to the decision-critical

Summary

Element	Value
Functional Requirements	FR ₁ : Facilitate legitimate biological research, FR ₂ : Prevent synthesis of Select Agents and Toxins
Design Parameters	DP ₁ : Order approval pipeline (justification review), DP ₂ : Pathogen screening module (sequence analysis + functional simulation)
Attack Vector	Fabricated research justification document inverting risk profile of synthesis order
Adversary Class	Ω_2 (Peripheral)
OODA Target Phase	Orient
Attack Pattern	FR Polarity Inversion